交接事宜

2026/04/08

Contents

- AIoTtalk+
- Kubernetes

AIoTtalk+ Overview

- AIoTtalk+ is a streaming-based multimedia IoT service platform
  - Integrates SIP semantics with IoTtalk
  - Supports the establishment of RTP-based multimedia streams

- Real-time Transport Protocol (RTP)
  - Designed for transmitting low-latency, real-time data over the Internet
  - Standardizes media encapsulation by adding timestamps and sequence numbers, enabling the receiver to reconstruct and synchronize the data stream

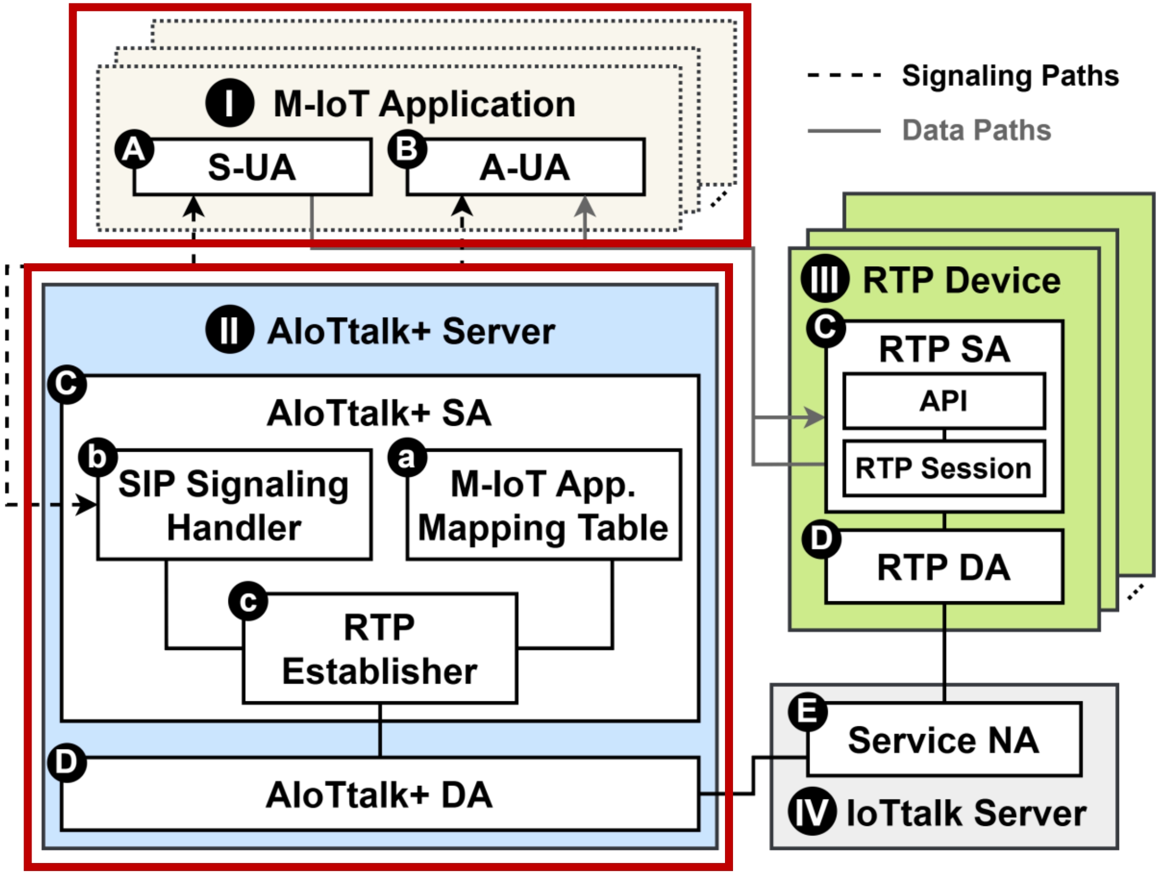
- Session Initiation Protocol (SIP)
  - Establishes, modifies, and terminates multimedia sessions
  - SIP messages include Session Description Protocol (SDP) payloads that convey media capabilities and transport parameters

AIoTtalk+ Overview

Architecture

- **M-IoT (Multimedia-IoT) Application**
  - Sensing User Agent (S-UA)
  - Actuating User Agent (A-UA)

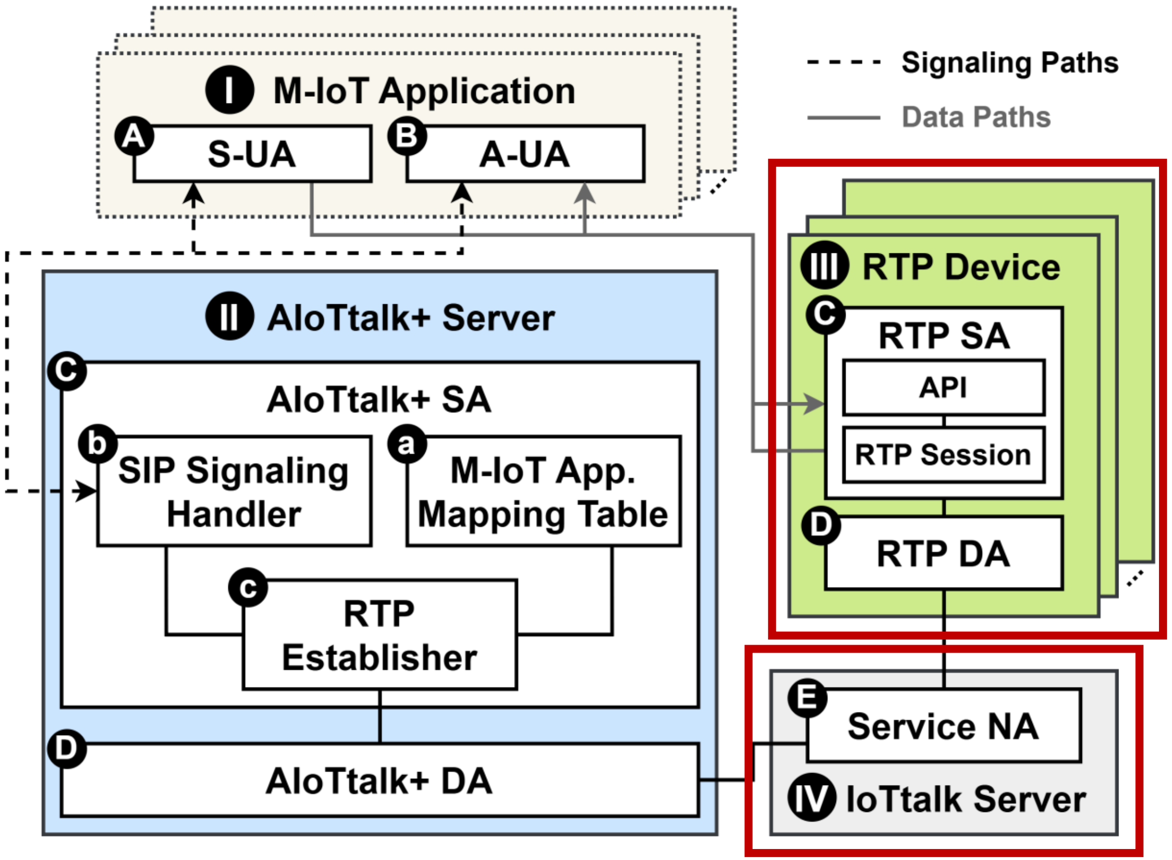
- **AIoTtalk+ Server**
  - AIoTtalk+ SA
    - M-IoT Application Mapping Table
    - SIP Signaling Handler
    - RTP Establisher
  - AIoTtalk+ DA



AIoTtalk+ Overview

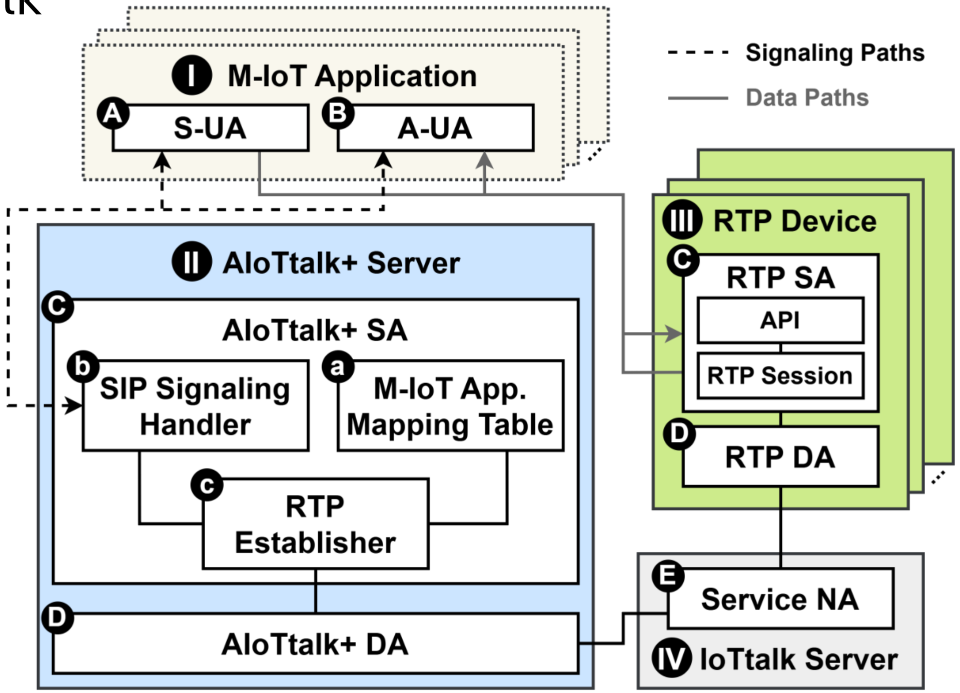
Architecture

- **RTP Device**
  - RTP SA
    - Application Programming Interface (API)
    - RTP Session
  - RTP DA
- **IoTtalk Server**
  - Service Network Application (NA)

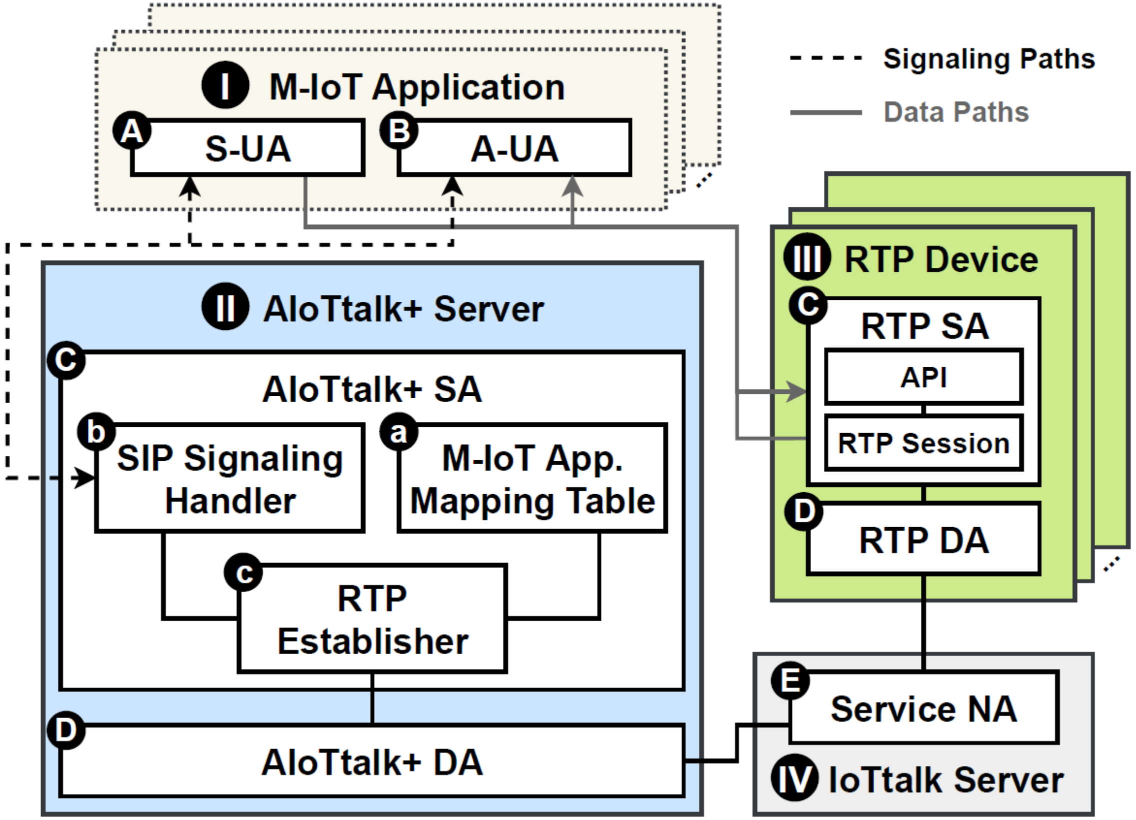


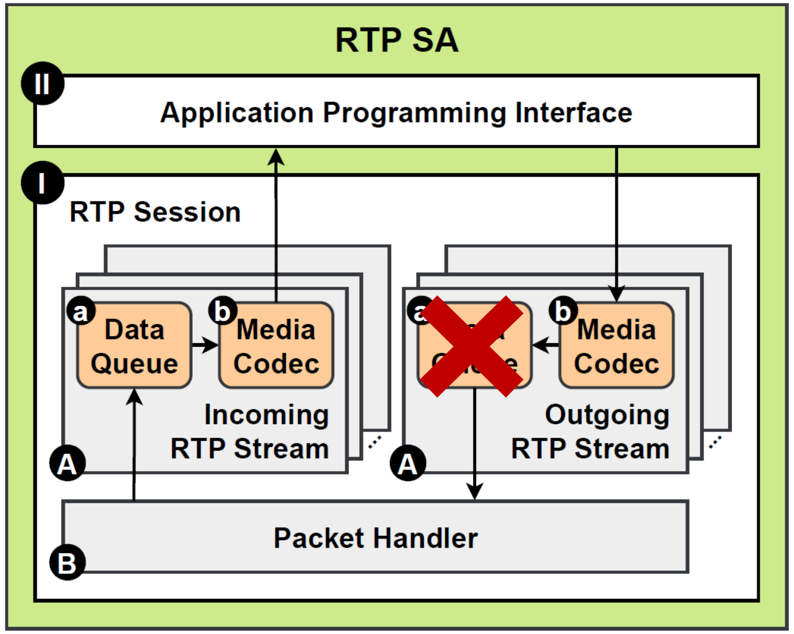
AIoTtalk+ Overview
Workflow

1. The S-UA/A-UA sends a SIP INVITE containing an SDP Offer to the SIP Signaling Handler.
2. The RTP Establisher parses media parameters and sends a corresponding Control Request to the service NA on the IoTtalk Server.
3. The RTP Device pulls the Control Request and initializes RTP streams accordingly, then pushes a Control Response containing IP address, port number and selected codec.
4. The RTP establisher retrieves the Control Response and generates SDP Answer accordingly.
5. The SIP Signaling Handler sends a 200 OK response with the SDP Answer in the message body to the S-UA/A-UA.
6. Successfully established RTP streams.



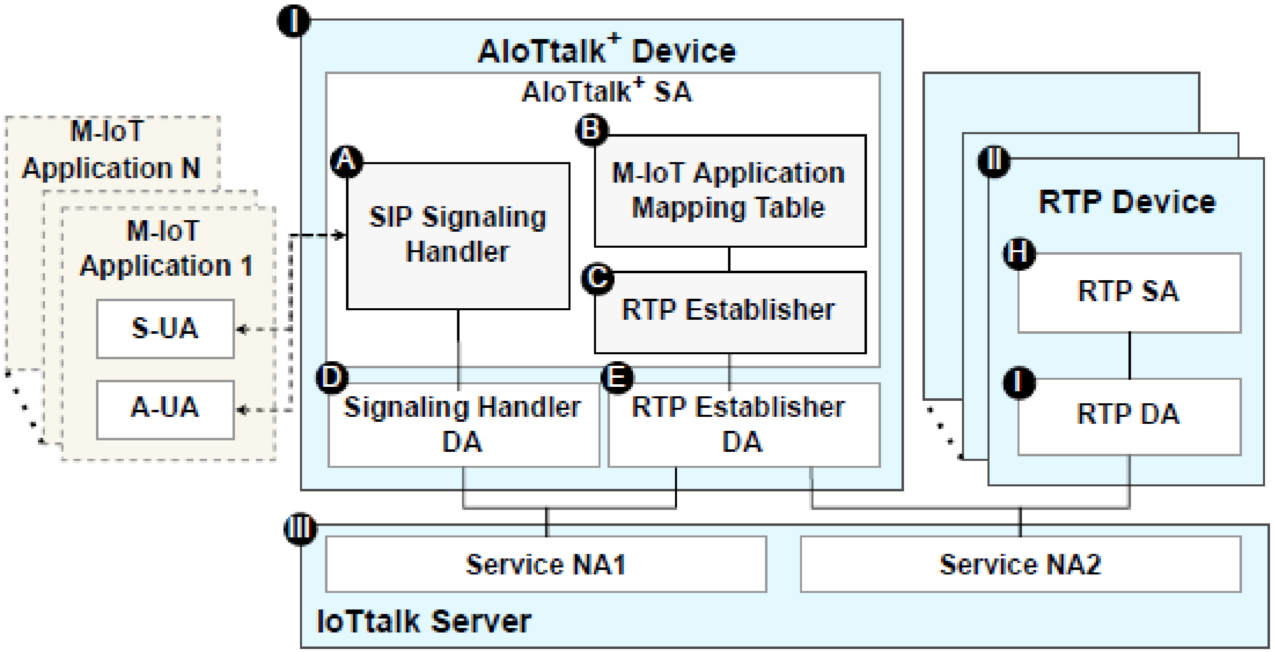
AIoTtalk+ 投稿版

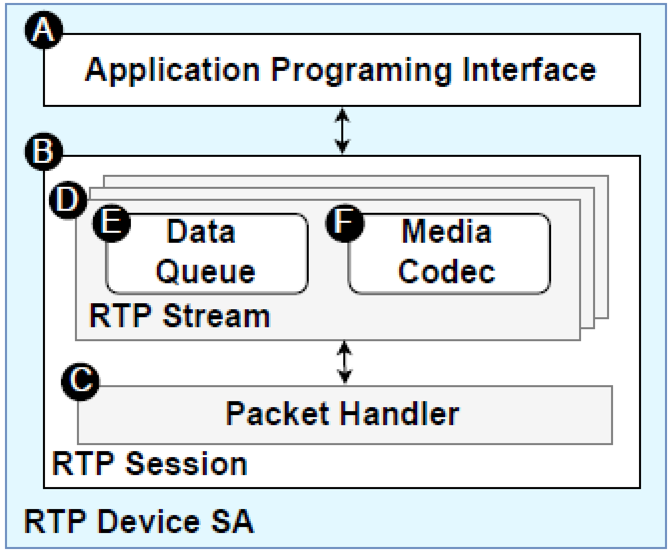


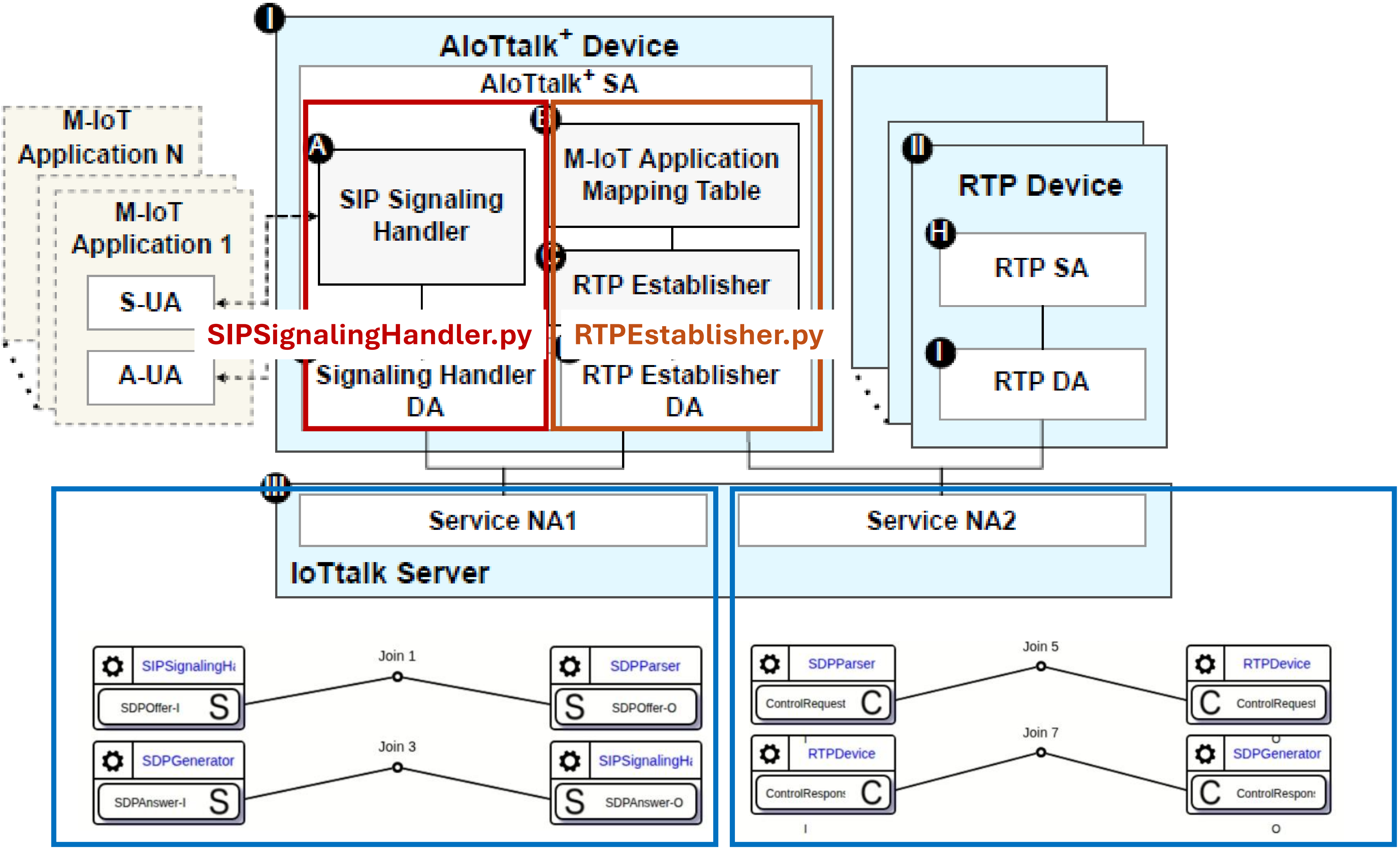


AIoTtalk+ 徐嘉呈學長碩論

- 2024MS_A_Streaming_Based_IoT_Service_Platform_and_Its_Application_To_UAV_Based_SLAM
- 接近code實作






AIoTtalk+ Device
AIoTtalk+ SA
M-IoT Application N
M-IoT Application 1
S-UA
A-UA
SIP Signaling Handler
M-IoT Application Mapping Table
RTP Establisher
SIPSignalingHandler.py
RTPEstablisher.py
Signaling Handler DA
RTP Establisher DA
RTP Device
RTP SA
RTP DA
Service NA1
Service NA2
IoTtalk Server
Join 1
Join 3
Join 5
Join 7
SDPParser
SDPOffer-I
SDPOffer-O
SDPGenerator
SDPAnswer-I
SDPAnswer-O
ControlRequest
RTPDevice

實際操作順序

- SIPSignalingHandler.py
- RTPEstablisher.py
- IoTtalk GUI
- RTPDevice
- SUA

要先開啟&註冊到M-IoT Application table
UA才有能對應的RTP device可以建立RTP連線

把M-IoT Application table改成動態更動的
RTP device包在pod裡面用K8s deploy

一些舊筆記

- RTPDevice & AIoTtalk+
  - https://hackmd.io/EsTA16a7RyqAIXoq8txH-Q
- RTP Device Trace Code
  - https://hackmd.io/4XAqIZm1SqCgH1gx3ih1ug

因為有些涉及環境

建議寫一份自己的操作順序&遇到的問題

Contents

- AIoTtalk+
- Kubernetes

Kubernetes (K8s)

Cluster 叢集

- Control Plane (Master Node)
- Node (Worker Node)
  - Pod

- 官方文件說明
  - https://kubernetes.io/docs/concepts/overview/components/
  - https://kubernetes.io/docs/concepts/architecture/
- 建置一個簡單的Cluster的筆記
  - https://hackmd.io/kGKP1bGMTsSTTIS1Gy8tMA